

## FUNCTIONAL REQUIREMENTS (What the system must do)

FR-01 Student Account Login The system shall allow students to create and log into an account within the application that is independent of UNT or campus authentication systems. **Isaiah**

### FR-02 Student Mobile App Enrollment and Check-In

The system shall provide a dedicated mobile application that allows students to (1) enroll face data by capturing one or more facial images using the device camera and (2) perform face-based attendance check-ins during class. **Moses**

FR-03 Multi-Angle Face Capture The system shall support multi-angle or guided face capture (e.g., head tilt or rotation) to improve face recognition accuracy. **Moses**

FR-04 Face Image Storage The system shall store enrolled face images in a project-controlled database separate from campus or Canvas databases. **Isaiah**

FR-05 Face Recognition Attendance Check-In The system shall allow students to scan their face during class to verify identity and record attendance. **Gabriel**

FR-06 Face Matching The system shall compare the in-class scan against stored face images to determine whether the student is present. **Austin**

FR-07 Automatic Attendance Marking The system shall automatically mark a student as present when a valid face match is detected. **Austin**

FR-08 Manual Attendance Override The system shall allow instructors to manually mark, change, or correct student attendance records. **Gabriel**

FR-09 Student Opt-Out Support The system shall allow students to opt out of face recognition attendance for privacy or personal reasons. **Cade**

FR-10 Opt-Out Attendance Handling The system shall route opted-out students to instructor-managed manual attendance tracking. **Cade**

FR-11 Instructor Dashboard The system shall provide an instructor-facing interface to view attendance records for a class session. **Isaiah**

FR-12 Attendance Status Indicators The system shall display attendance states (e.g., present, absent, late, flagged) in the instructor interface. **Isaiah**

FR-13 Configurable Grace Period The system shall allow instructors to configure a grace period for late arrivals. **Austin**

FR-14 Late Arrival Detection The system shall detect and record when a student checks in after the configured grace period. **Cade**

FR-15 Extended Absence Flagging The system shall flag students who leave or are inactive beyond a configurable time threshold and notify the instructor for review. **Moses**

FR-16 No Automatic Penalties for Extended Absence The system shall not automatically mark students absent for extended absences and shall defer final decisions to the instructor. **Gabriel**

FR-17 Real-Time Attendance Processing The system shall record attendance in real time at the beginning of class. **Austin**

FR-18 Notification of Attendance Status The system shall notify students when they are marked absent or when attendance fails. **Austin**

FR-19 Face Recognition Error Feedback The system shall provide clear error messages for failed face scans (e.g., face not detected, poor lighting). **Cade**

FR-20 Face Data Update The system shall allow students to update or re-scan their face data when recognition issues occur. **Gabriel**

FR-21 QR Code Onboarding The system shall support QR code-based onboarding to guide students to the app download or attendance page. **Cade**

FR-22 Instructor Attendance Review by Date The system shall allow instructors to view attendance records by date (e.g., calendar-based view). **Austin**

FR-23 Attendance Data Export The system shall allow instructors to export attendance data in CSV or PDF format. **Austin**

FR-24 Canvas API Integration (Optional) The system shall support optional integration with Canvas via the Canvas API for attendance data transfer. **Isaiah**

FR-25 Offline Attendance Fallback The system shall support instructor-managed attendance when internet connectivity is unavailable. **Moses**

## NON-FUNCTIONAL REQUIREMENTS (How the system must behave)

### NFR-01 Privacy Compliance

The system shall not access or store campus ID photos or institutional identity data. **Cade**

### NFR-02 Data Separation

All student face and attendance data shall be stored only in project-controlled databases. **Cade**

NFR-03 Instructor Authority

The system shall always allow instructors to override automated attendance decisions.**Gabriel**

NFR-04 Minimal Instructor Setup **Isaiah**

The system shall require minimal setup and no administrative privileges on instructor devices.

NFR-05 Platform Accessibility **Isaiah**

The system shall be accessible via standard web browsers for instructors.

NFR-06 Mobile Compatibility

The system shall function on camera-enabled mobile devices for students.**Gabriel**

NFR-07 Scalability

The system shall support classes ranging from small sections to large lectures (100+ students).**Moses**

NFR-08 Reliability

The system shall handle face recognition failures gracefully without penalizing students automatically.**Moses**

NFR-09 Usability

The instructor interface shall be simple and intuitive, similar to existing attendance tools.**Moses**

NFR-10 Cost Constraints

The system shall rely on free or low-cost infrastructure suitable for a prototype.**Gabriel**